

YINCHENG ZHOU

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EDUCATION

University of Pennsylvania

September 2025 - Present

Vagelos Integrated Program in Energy Research (VIPER)

Philadelphia, PA

- Dual Degree: B.S.E. in Computer Science (AI Concentration) | B.A. in Physics & Astronomy
- Relevant Coursework: Big Data Analysis, Computer Systems, Data Structures & Algorithms
- GPA: 4.0 / 4.0

EXPERIENCE

Franklink, Inc. | Founding Software Engineer [Demo], Philadelphia, PA

October 2025 - February 2026

- Built an iMessage-based networking assistant, automating matchmaking via multi-agent orchestration.
- Implemented persisted agent memory and selective dispatch across 5 sub-agents; cut user-facing latency by 30%.
- Designed agentic RAG with permissioned access across 8 tools (DB, email, calendar) for context-aware retrieval and recommendation.
- Engineered a Kafka-backed event pipeline for LLM-heavy messaging with at-least-once delivery, idempotency keys, and retry topics (backoff) + DLQ; supported 50+ concurrent sessions while keeping p99 end-to-end latency within 40% of single-session baseline.
- Containerize services and design test/deploy workflows; ship on AWS ECS.

GRASP Laboratory | ML Researcher, Advisor: Prof. Chaudhari, Philadelphia, PA

January 2026 - Present

- Designed SPD, a sparse depth prediction algorithm that decodes only at queried pixels, achieving $2.9\times$ inference speedup over dense baselines for real-time robotics.
- Implemented a lightweight per-query decoder (11M params) with multi-scale cross-attention, cross-query communication, and deformable sampling, enabling 72 Hz inference at 256 query points with 544×416 resolution on RTX 4060.
- Architected dense canvas supervision strategy to address sparse query information loss during training, enabling convergence on NYU Depth V2 without ground truth densification.

Human Machine Intelligence Lab, Peking University | ML Researcher, Remote

March 2026 - Present

- Architected action-level MCTS SFT for GUI agent training with step-level data collection and difficulty-weighted loss, outperforming agentic RL (GRPO) with only 25% computational cost; boosted OSWorld benchmark from 12.3% to 17.7% (+44% relative).
- Built a fault-tolerant distributed rollout system across 112 Docker VMs and 8 A100 GPUs with self-healing monitoring and action-replay VM cloning; collected 2,927 trajectories and audited 35K steps via 200 parallel LLM agents for step-level quality masking.

PROJECTS

AlphaOne | Subject oriented financial sentiment platform [GitHub] [Demo] [HuggingFace]

October 2025 - Present

- Fine-tuned DeBERTa to classify sentiment per ticker in multi-entity financial text; achieving 82.5% accuracy (0.823 Macro F1), increasing performance by 65 percentage points from FinBERT baseline.
- Curated 6,287 training triples: scraped 7 subreddits, LLM-labeled via Ollama, hand-audited 4,501 pairs (615 corrections).
- Architected a 6-layered sentiment pipeline with idempotent content hashing for concurrent-safe batch ingestion.
- Engineered concurrent-safe, idempotent batch ingestion with content versioning; deployed via Docker Compose.

NL2SQL Bot | [GitHub] [Demo]

December 2025 - January 2026

- Built a multi-agent NL2SQL system on Google ADK converting natural language to validated SQL with interpretive visualization.
- Orchestrated 5 agents with session state passing, loop-guarded retry, and per-agent LLM overrides for cost-performance tuning.
- Enforced read-only execution via 18 blocked SQL patterns and table allowlisting; abstracted dialect rules across MySQL, PostgreSQL, and SQLite.

SKILLS & ACHIEVEMENTS

Languages

C++, Python, Java, JavaScript/TypeScript, SQL, Assembly, OCaml

Backend & DevTools

FastAPI, Spring Boot, Kafka, Redis, Celery, Docker, AWS, Git

ML & Data

PyTorch, CUDA, spaCy, Pandas, LangGraph, Google ADK

Achievements

Team China, International Science and Engineering Fair (ISEF)

Silver Medal, Team China, International Young Physicists' Tournament (IYPT)

Second Prize, China Mathematics Olympiad

Second Prize, China Physics Olympiad, Second Prize

Champion, Start-up in a Weekend Hackathon, Photon Hybrid Intelligence Track